

CALVINIA, South Africa

WMO: 686180

Lat: 31.4819S

Lon: 19.7617E

Elev: 3205

StdP: 13.07

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	32.1	9.4	10.0	67.3	12.8	11.8	65.9	27.5	54.4	24.9	55.4	5.7	80	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.3	96.2	61.0	93.5	60.4	91.2	59.8	65.7	82.9	64.4	83.0	63.0	82.2	12.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	90.0	70.7	58.5	82.5	70.1	56.2	75.9	69.7	32.3	82.2	31.2	82.9	30.1	81.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.4	20.9	18.7	DB	25.6	101.4	1.9	2.0	24.3	102.8	23.1	104.0	22.1	105.1	20.7	106.5	(4)
(5)			WB	23.8	68.4	2.2	1.8	22.2	69.7	20.9	70.7	19.6	71.7	18.0	73.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	73.2	73.4	69.7	63.4	56.8	51.2	50.6	51.1	55.7	62.0	65.3	69.8		
	DBStd	10.77	6.34	5.97	6.35	7.37	7.15	6.36	7.13	7.25	7.39	7.90	7.25	6.16		
	HDD50	262	0	0	0	2	18	59	80	74	24	4	1	0		
	HDD65	2277	9	5	26	116	268	416	450	434	293	155	82	23		
	CDD50	4566	719	654	612	405	230	95	98	109	195	375	461	613		
	CDD65	1105	264	239	172	69	15	1	3	3	14	62	92	171		
	CDH74	15867	3595	3113	2276	925	217	10	13	93	352	1165	1574	2534		
	CDH80	6942	1782	1516	978	299	28	0	0	14	90	460	645	1130		
(14) Wind	WSAvg	9.1	9.4	9.0	8.4	8.7	8.4	9.1	9.6	9.3	8.9	9.6	9.7	9.4		
(15) Precipitation	PrecAvg	8.0	0.3	0.4	0.9	0.9	0.9	1.2	0.9	0.8	0.5	0.4	0.5	0.4		
	PrecMax	18.4	2.5	3.0	3.9	4.7	3.1	4.7	3.7	2.1	2.8	2.3	3.4	2.2		
	PrecMin	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.8	0.6	0.6	0.9	1.0	0.8	0.9	0.9	0.6	0.6	0.5	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.6	99.5	96.2	91.1	84.0	75.4	75.7	81.5	87.8	94.0	96.0	97.0		
		MCWB	62.2	62.4	61.3	60.0	55.9	51.4	51.6	52.1	55.1	58.8	58.8	60.1		
	2%	DB	96.0	95.4	92.2	86.9	79.6	72.2	72.2	76.3	82.8	89.2	90.7	93.1		
		MCWB	61.2	61.6	60.6	58.6	54.0	50.3	50.2	51.0	53.9	57.3	57.6	59.4		
	5%	DB	92.7	92.4	89.3	83.6	76.3	69.5	69.2	72.0	78.5	85.7	87.0	90.0		
		MCWB	60.6	61.2	59.2	57.2	53.2	49.2	48.8	49.7	52.3	56.0	56.6	58.8		
	10%	DB	89.6	89.4	85.9	79.7	72.7	66.1	65.8	67.8	73.6	80.9	83.2	86.8		
		MCWB	59.5	60.5	58.3	56.2	52.0	48.1	47.7	48.0	50.6	54.3	55.5	58.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	68.2	66.4	63.5	58.4	54.4	54.1	55.7	59.0	61.9	63.6	65.3		
		MCDB	81.9	83.1	82.3	80.3	73.7	67.6	70.8	70.4	76.8	84.3	83.7	82.7		
	2%	WB	65.2	66.0	64.6	61.7	56.4	52.3	51.6	52.9	56.0	59.6	61.0	63.2		
		MCDB	84.1	83.0	80.8	78.3	73.3	65.7	67.4	70.8	76.0	83.2	79.3	82.7		
	5%	WB	63.7	64.5	62.9	60.1	54.9	50.8	50.0	51.0	53.8	57.7	58.8	61.7		
		MCDB	84.4	82.7	79.1	77.5	71.2	64.2	65.3	68.2	74.5	81.0	79.6	82.2		
	10%	WB	62.1	62.9	61.3	58.2	53.4	49.4	48.5	49.1	51.7	55.7	57.1	60.2		
		MCDB	83.2	83.0	78.5	75.3	68.6	62.1	62.6	65.2	70.6	77.8	78.7	80.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	29.3	29.0	28.6	26.0	23.9	22.4	22.9	24.7	27.6	29.7	29.0	29.4		
		MCDBR	31.6	31.9	32.6	31.5	29.3	27.8	27.5	31.7	35.4	36.2	34.5	33.6		
		MCWBR	10.9	12.0	12.3	12.6	12.2	12.6	12.6	13.8	14.7	15.1	13.0	12.3		
	5% WB	MCDBR	26.3	25.5	25.1	26.2	24.1	22.7	23.8	28.0	30.7	31.2	30.2	28.4		
		MCWBR	9.4	9.7	10.4	10.9	10.8	11.7	11.9	13.1	13.6	13.6	12.0	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.303	0.297	0.283	0.269	0.259	0.262	0.281	0.293	0.305	0.288	0.291		
	taud		2.617	2.603	2.612	2.665	2.682	2.672	2.646	2.557	2.489	2.490	2.609	2.621		
	Ebn at Noon		331	323	315	304	294	290	293	299	311	319	332	334		
	Edh at Noon		32	32	30	25	22	21	22	27	33	35	32	32		
(44) All-Sky Solar Radiation	RadAvg		2739	2476	2051	1546	1167	985	1090	1363	1818	2279	2628	2822		
	RadStd		76	75	65	75	67	57	48	63	80	84	97	112		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.87	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.87	N/A	N/A	N/A	N/A

Nomenclature: See separate page